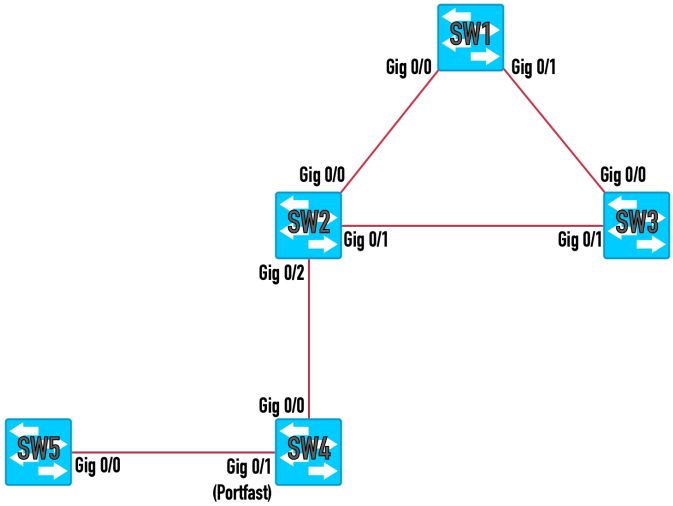
BPDU Guard

Topology



Initial Configuration Commands

SW1:

```
en
conf t
no ip domain-lookup
line con 0
logging synchronous
exec-timeout 0 0
hostname SW1
spanning-tree mode rapid-pvst
spanning-tree extend system-id
spanning-tree vlan 1 priority 24576
end
copy run star
```

SW2:

```
en
conf t
no ip domain-lookup
line con 0
logging synchronous
```

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```
exec-timeout 0 0
hostname SW2
spanning-tree mode rapid-pvst
spanning-tree extend system-id
end
copy run star
```

SW3:

```
en
conf t
no ip domain-lookup
line con 0
logging synchronous
exec-timeout 0 0
hostname SW3
spanning-tree mode rapid-pvst
spanning-tree extend system-id
end
copy run star
```

SW4:

```
en
conf t
no ip domain-lookup
line con 0
logging synchronous
exec-timeout 0 0
hostname SW4
spanning-tree mode rapid-pvst
spanning-tree extend system-id
int gig 0/1
spanning-tree portfast edge
end
copy run star
```

SW5:

```
en
conf t
```

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```
no ip domain-lookup
line con 0
logging synchronous
exec-timeout 0 0
hostname SW5
spanning-tree mode rapid-pvst
spanning-tree extend system-id
int gig 0/0
shutdown
end
copy run star
```

Lab Tasks

1. First, let's confirm that Gig 0/1 on SW4 is configured for PortFast by using the "show spanning-tree" command.
2. On SW5, Gig 0/0 is administratively shutdown. Let's go ahead and bring that interface up. This will cause a BPDU to be sent into Gig 0/1 on SW4. Let's confirm this by using the "show spanning-tree" command on SW4.
3. On SW4, we need to re-enable PortFast on Gig 0/1. On SW5, let's shutdown Gig 0/0. On SW4, shutdown Gig 0/1 and then turn it back on. Finally, verify that Gig 0/1 is functioning as a PortFast port again.
4. First, let's enable BPDU Guard on interface Gig 0/1, and then remove it. After that, enable BPDU Guard globally.
5. On SW4, check that BPDU Guard is enabled globally by using the command "show spanning-tree summary."
6. Go back to SW5 and enable Gig 0/0. Whenever we enable this interface, it is going to send out a BPDU.
7. On SW4, let's put in the command "show interface gig 0/1" to see if Gig 0/1 has gone into an err-disabled state.

Solution

Step 1: First, let's confirm that Gig 0/1 on SW4 is configured for PortFast by using the "show spanning-tree" command.

```
SW4>en
SW4#show spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    24577
```

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```
             Address     5254.0013.0e89
             Cost        8
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     5254.0009.f309
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
----------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p Edge
Gi0/2               Desg FWD 4         128.3    P2p
Gi0/3               Desg FWD 4         128.4    P2p
```

(##The "Edge" indicates that we are configured for PortFast on Gig 0/1.)

Step 2: On SW5, Gig 0/0 is administratively shutdown. Let's go ahead and bring that interface up. This will cause a BPDU to be sent into Gig 0/1 on SW4. Let's confirm this by using the "show spanning-tree" command on SW4.

```
SW5>en
SW5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW5(config)#int gig 0/0
SW5(config-if)#no shutdown

SW4#show spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    24577
             Address     5254.0013.0e89
             Cost        8
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     5254.0009.f309
```

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```
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
----------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p
Gi0/2               Desg FWD 4         128.3    P2p
Gi0/3               Desg FWD 4         128.4    P2p
```

(##You can see that this "Edge" parameter is now gone. Gig 0/1 is no longer acting as a PortFast port, it is acting as a regular switch port.)

Step 3: On SW4, we need to re-enable PortFast on Gig 0/1. On SW5, let's shutdown Gig 0/0. On SW4, shutdown Gig 0/1 and then turn it back on. Finally, verify that Gig 0/1 is functioning as a PortFast port again.

```
SW5(config-if)#shutdown

SW4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW4(config)#int gig 0/1
SW4(config-if)#shutdown
SW4(config-if)#no shutdown
*Aug 17 20:52:49.876: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed
state to administratively down
*Aug 17 20:52:50.876: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to down
SW4(config-if)#no shutdown
SW4(config-if)#end

*Aug 17 20:52:53.680: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed
state to up
*Aug 17 20:52:54.681: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
SW4#
*Aug 17 20:52:55.091: %SYS-5-CONFIG_I: Configured from console by console
SW4#show spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    24577
             Address     5254.0013.0e89
             Cost        8
             Port        1 (GigabitEthernet0/0)
```

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```
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     5254.0009.f309
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ------------------------------
---
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p Edge
Gi0/2               Desg FWD 4         128.3    P2p
Gi0/3               Desg FWD 4         128.4    P2p
```

(##Here, you can see that Gig0/1 on SW4 is now a PortFast port again.)

Step 4: First, let's enable BPDU Guard on interface Gig 0/1, and then remove it. After that, enable BPDU Guard globally.

```
SW4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW4(config)#int gig 0/1
SW4(config-if)#spanning-tree bpduguard enable
SW4(config-if)#no spanning-tree bpduguard enable
SW4(config-if)#exit
SW4(config)#spanning-tree portfast edge bpduguard default
SW4(config)#end
```

(##BPDU Guard will be configured on any non-trunking port that is configured for PortFast whenever we globally configure BPDU Guard.)

Step 5: On SW4, check that BPDU Guard is enabled globally by using the command "show spanning-tree summary."

```
SW4#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: none
Extended system ID                       is enabled
Portfast Default                         is disabled
Portfast Edge BPDU Guard Default         is enabled
Portfast Edge BPDU Filter Default        is disabled
Loopguard Default                        is disabled
```

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```
PVST Simulation Default                 is enabled but inactive in rapid-pvst
mode
Bridge Assurance                        is enabled
EtherChannel misconfig guard            is enabled
Configured Pathcost method used is short
UplinkFast                              is disabled
BackboneFast                            is disabled

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                      0         0        0          4          4
---------------------- -------- --------- -------- ---------- ----------
1 vlan                        0         0        0          4          4
```

(##Here we can see that BPDU Guard has been enabled globally.)

Step 6: Go back to SW5 and enable Gig 0/0. Whenever we enable this interface, it is going to send out a BPDU.

```
SW5(config-if)#no shutdown
SW5(config-if)#end
```

Step 7: On SW4, let's put in the command "show interface gig 0/1" to see if Gig 0/1 has gone into an err-disabled state.

```
*Aug 17 20:58:14.326: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port
GigabitEthernet0/1 with BPDU Guard enabled. Disabling port.
*Aug 17 20:58:14.326: %PM-4-ERR_DISABLE: bpduguard error detected on Gi0/1,
putting Gi0/1 in err-disable state
*Aug 17 20:58:15.329: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to down
*Aug 17 20:58:16.328: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed
state to down
SW4#show int gig 0/1
GigabitEthernet0/1 is down, line protocol is down (err-disabled)
  Hardware is iGbE, address is 5254.0015.1e70 (bia 5254.0015.1e70)
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Auto Duplex, Auto Speed, link type is auto, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:55, output 00:00:56, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
```

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```
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     25 packets input, 4497 bytes, 0 no buffer
     Received 24 broadcasts (24 multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 watchdog, 24 multicast, 0 pause input
     735 packets output, 59076 bytes, 0 underruns
     0 output errors, 0 collisions, 3 interface resets
     0 unknown protocol drops
     0 babbles, 0 late collision, 0 deferred
     1 lost carrier, 0 no carrier, 0 pause output
     0 output buffer failures, 0 output buffers swapped out
```

(##With this syslog message, we can already see that BPDU Guard is kicking in by changing the state of Gig 0/1 to down.)
(##We can see that Gig 0/1 is down, the line protocol is down, and the reason this is is because we are in an err-disabled state.)
(##BPDU Guard can help prevent a Layer 2 topological loop from forming due to a switch being connected to a port configured for PortFast. Specifically, a port configured for PortFast bypasses the STP Listening and Learning states and immediately transitions to the Forwarding state, which negates the loop prevention offered by STP.)

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